

XRPL Validator — Daily Engineering Report • 2026-07-27

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **8 commits** (xrpl-validator 8)

Engineering changes

- **fix(xrpl-ledger): reap a book level's dead offers before consulting the pool** (01f09ba , xrpl-validator) rippled steps the offer stream BEFORE it asks the AMM, and every dead offer stepped past is removed whether or not anything crossed: `if (offers.step()) // BookStep.cpp:835 if (tryAMM(offers.tip().quality())) // anchor = first LIVE offer do { ... } while (offers.step()); else tryAMM(std::nullopt); // exhausted -> maxOffer return {offers.permToRemove(), ...}; // BookStep.cpp:852`
- **fix(xrpl-ledger): step past a satisfied fill, and reap dust-backed offers** (130c246 , xrpl-validator) Two more of `OfferStream::step`'s removal semantics, both needed to close #105778999 6B2A11B3 (5v15, missing 10, extra 0) -- a circular self-pay, SendMax 140 XRP into POSAA, `tfPartialPayment`.
- **fix(xrpl-ledger): carry the trailing reap across book levels** (ba73d4f , xrpl-validator) The trailing step added in 130c246 stopped at the end of the page chain of the level it had been crossing on. rippled's stream spans the whole BOOK: `OfferStream::step` carries no quality test at all, so after a consumed offer it keeps advancing — and reaping — across quality levels until it reaches a LIVE offer, and only then does `execOffer`'s `checkQualityThreshold` return false and end the walk ...
- **fix(xrpl-ledger): a crossing that exhausts the account rests no remainder** (bb522bf , xrpl-validator) "If offer crossing exhausted the account's funds don't create the offer" (`OfferCreate.cpp:479-484`). This is the companion to the pre-crossing `accountFunds` clamp from 1de6d87: that one bounds what may be SPENT, this one bounds what may REST.
- **fix(xrpl-ledger): tick re-derivation divides with the STAmount rule** (1edaa63 , xrpl-validator) #105777146 9BA91A9E sells 0.00059094 WETH for 1 XRP. The WETH issuer publishes TickSize 6, so the rate rounds up to 6 significant digits and the non-exact side is re-derived before placing. We derived `TakerGets` 0.000590939712330548 where mainnet stored 0.0005909397123305481 -- one ulp higher -- and that single ulp moved the rested offer to book level ...ABFB5801 where mainnet wrote ...ABFB5800. 4v4, the ledger's ...
- **fix(xrpl-ledger): a bridged pass needs its book composition within the limit** (625399c , xrpl-validator) #105807256 84FD7DC8, the last divergence in the harness. A `tfSell|tfPassive OfferCreate` selling 3105.662739 BBRL for 607.478737 RLUSD: mainnet rests it in 4 nodes, we crossed 8 objects (12v4, all extra, nothing missing).
- **fix(xrpl-ledger): a frozen holder can fund nothing** (1c46cda , xrpl-validator) rippled reads spendable IOU through `accountHolds` with `fhZERO_IF_FROZEN`, so a holder whose line the issuer has frozen has NO funds at all, however large the balance. `isFrozen` (`RippleStateHelpers.cpp:127`) is the issuer's GLOBAL freeze, or the

ISSUER's side of the line -- `(issuer > account) ? lsfHighFreeze : lsfLowFreeze`. Our `available()` read the balance and nothing else, so a frozen holder could trade freely.

Docs and operations

- **docs(xrpl-ledger): correct the stale directory.rs scope note** (`51f6fef` , xrpl-validator)

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **WASM-Gated Escrow Finish Predicates** (hooks) Add flag `tfWasmCondition` and `SField sfFinishPredicate` (a size-capped, $\leq 4\text{KB}$ WASM blob run by a deterministic bounded interpreter — fixed step ceiling,
- **Threshold-Delayed Outbound Payment Holds** (custody) Add `AccountPolicySet` writing an `AccountPolicy SLE` (owner-reserved, `keylet::accountPolicy(account)`) with `sfPolicyThreshold`, `sfHoldDelay` (ledgers), and
- **Snapshot-Based MPT Corporate Actions** (rwa) Add issuer-only `CorporateActionCreate`, which creates an `ltCORPORATE_ACTION` object (owner-reserved, `keylet::corpAction(issuer, sequence)`) referencing a

Appendix — commits

Time	Repo	Commit	Subject
13:06	xrpl-validator	<code>01f09ba</code>	fix(xrpl-ledger): reap a book level's dead offers before consulting the pool
13:07	xrpl-validator	<code>51f6fef</code>	docs(xrpl-ledger): correct the stale directory.rs scope note
19:22	xrpl-validator	<code>130c246</code>	fix(xrpl-ledger): step past a satisfied fill, and reap dust-backed offers
19:45	xrpl-validator	<code>ba73d4f</code>	fix(xrpl-ledger): carry the trailing reap across book levels
20:22	xrpl-validator	<code>bb522bf</code>	fix(xrpl-ledger): a crossing that exhausts the account rests no remainder
21:15	xrpl-validator	<code>1edaa63</code>	fix(xrpl-ledger): tick re-derivation divides with the <code>STAmount</code> rule
21:50	xrpl-validator	<code>625399c</code>	fix(xrpl-ledger): a bridged pass needs its book composition within the limit
23:18	xrpl-validator	<code>1c46cda</code>	fix(xrpl-ledger): a frozen holder can fund nothing